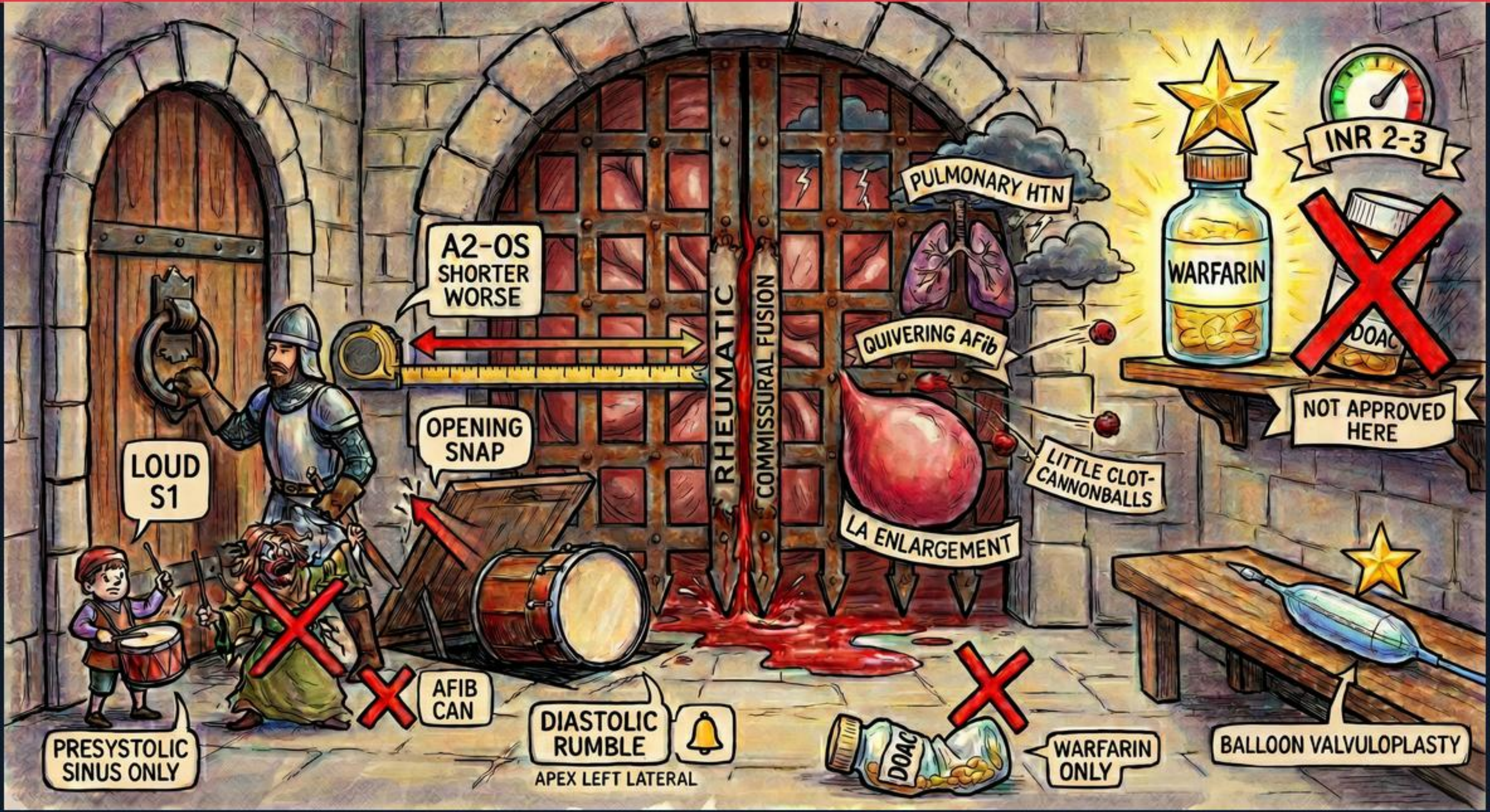
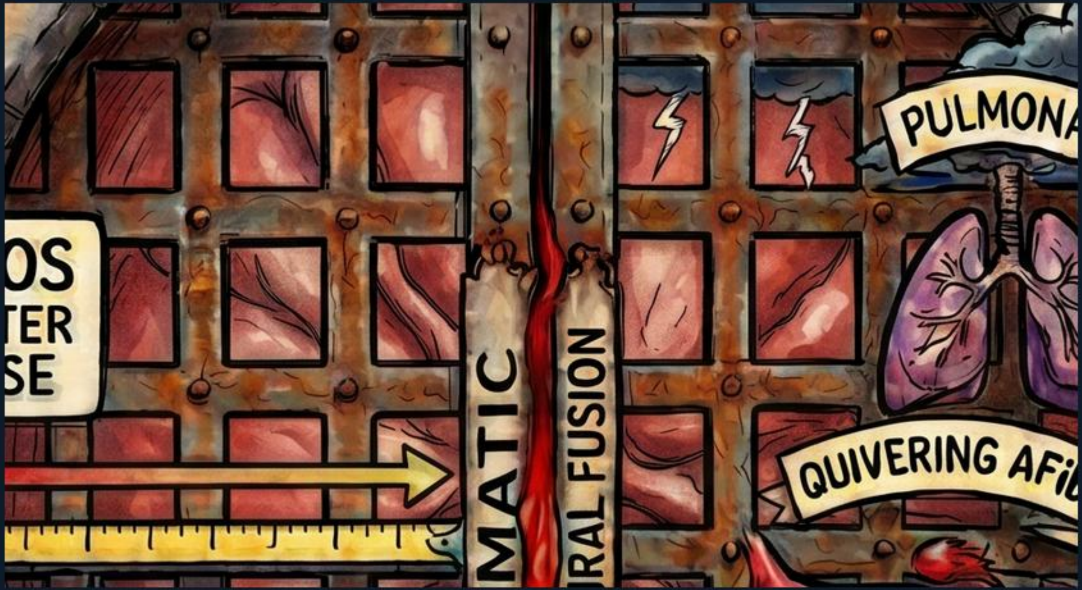
Valvular — Mitral Stenosis



14 symbols · sketch walkthrough



Rheumatic / commissural fusion

WHAT YOU SEE

The huge wooden castle gate labeled RHEUMATIC with COMMISSURAL FUSION written down the central seam where the doors are fused shut.

WHAT IT ENCODES

Rheumatic heart disease is the dominant cause of mitral stenosis worldwide. The mechanism is post-inflammatory fusion of the valve commissures, plus leaflet thickening and chordal shortening, which narrows the mitral orifice. The fused doors picture a valve that can no longer open fully in diastole.

THE BOARD TRAP

Calling degenerative/calcific disease the classic cause of significant MS — the board answer is rheumatic commissural fusion.

Loud S1

WHAT YOU SEE

The knight ringing the heavy iron door-knocker, captioned LOUD S1.

WHAT IT ENCODES

In MS the mitral leaflets stay wide open until forced shut, so closure is abrupt and S1 is loud — as long as the leaflets remain pliable. A loud S1 is an early auscultatory clue to mobile, non-calcified rheumatic valve.

THE BOARD TRAP

Assuming S1 stays loud in advanced MS — once leaflets calcify and become immobile, S1 actually softens.

Opening snap

WHAT YOU SEE

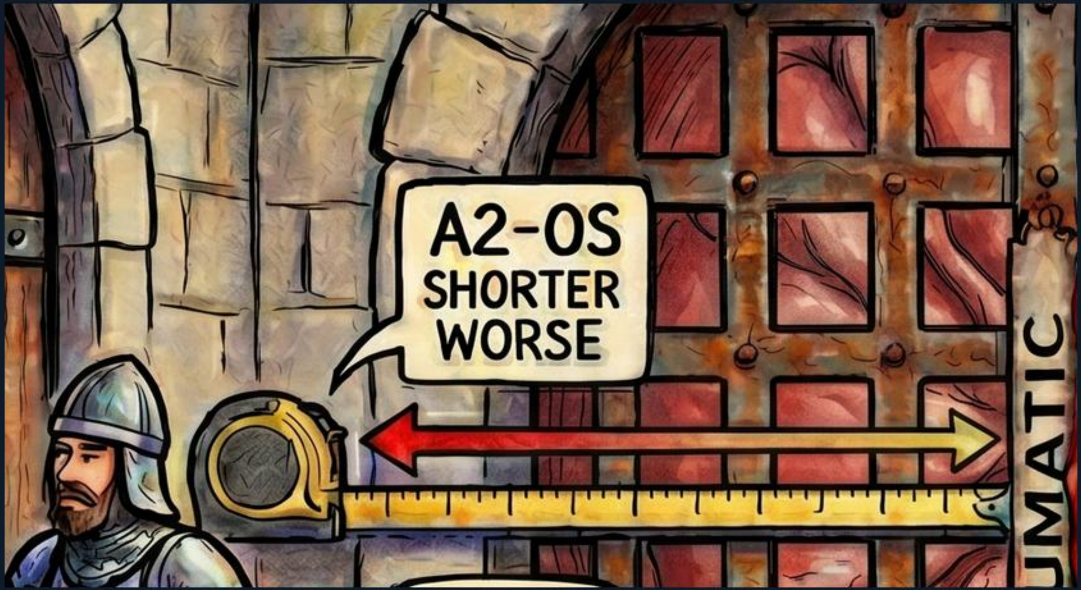
The drawbridge plank swinging down, labeled OPENING SNAP.

WHAT IT ENCODES

After S2, the stenotic mitral valve domes open with a high-pitched OPENING SNAP early in diastole. It marks the start of the diastolic rumble and signals a still-pliable valve. The snap disappears when the valve becomes heavily calcified and rigid.

THE BOARD TRAP

Confusing the opening snap with an S3 — the OS is high-pitched and earlier; an S3 would actually be unusual in pure MS.



A2–OS interval: shorter = worse

WHAT YOU SEE

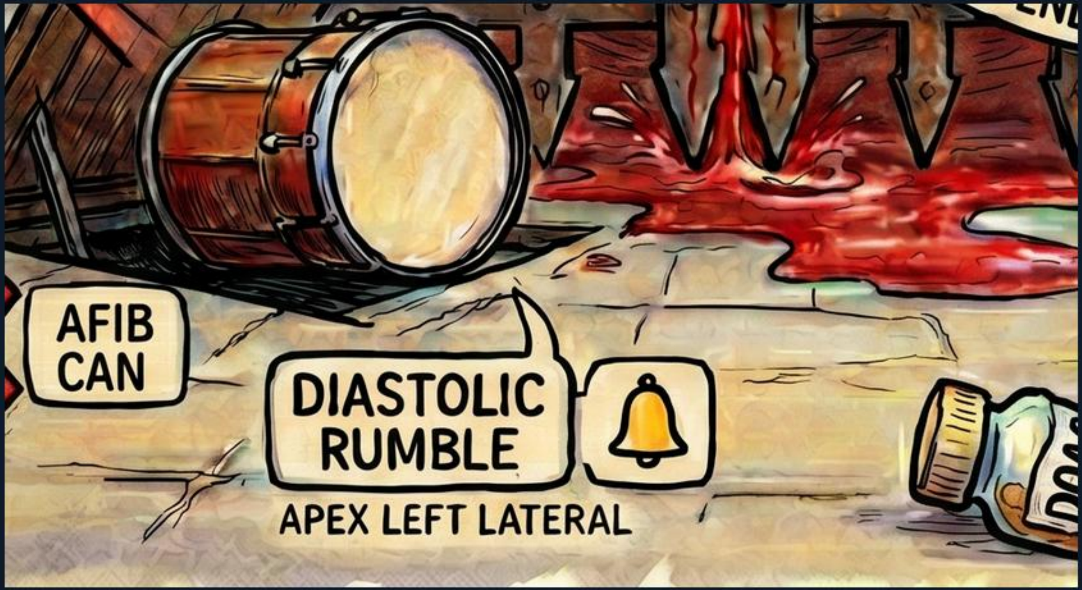
A measuring tape / spear shortening with an arrow, captioned A2-OS SHORTER WORSE.

WHAT IT ENCODES

The interval between A2 and the opening snap is the single best auscultatory severity clue. Higher left atrial pressure forces the valve open sooner, so the snap moves closer to S2. A SHORT A2-OS interval means more severe stenosis; a long interval means milder disease.

THE BOARD TRAP

Thinking a longer A2-OS interval means worse MS — it is the reverse: shorter interval = higher LA pressure = more severe.



Diastolic rumble at apex

WHAT YOU SEE

The kettledrum on the ground labeled DIASTOLIC RUMBLE with a bell icon and APEX LEFT LATERAL.

WHAT IT ENCODES

The murmur of MS is a low-pitched mid-diastolic rumble heard best at the apex with the bell, patient in the left lateral decubitus position. It is low-frequency turbulent flow across the narrowed valve — easy to miss unless you position and listen with the bell.

THE BOARD TRAP

Listening with the diaphragm in the sitting patient — the low-pitched rumble needs the BELL at the apex in left lateral decubitus.

Presystolic accentuation — sinus only

WHAT YOU SEE

The small child beating a drum, captioned PRESYSTOLIC SINUS ONLY.

WHAT IT ENCODES

In sinus rhythm, atrial contraction pushes extra blood across the valve just before S1, producing a presystolic accentuation of the rumble. This phase depends on an effective atrial kick.

THE BOARD TRAP

Expecting presystolic accentuation in atrial fibrillation — without organized atrial contraction it is lost (the next symbol).

Presystolic murmur lost in AFib

WHAT YOU SEE

A red X over the drumming figure, captioned AFIB CAN (cancels it).

WHAT IT ENCODES

When MS patients go into atrial fibrillation, the atrial kick is gone, so the presystolic accentuation of the murmur disappears. Recognizing this prevents you from doubting the diagnosis when the presystolic component is absent in an irregular rhythm.

THE BOARD TRAP

Calling a missing presystolic murmur evidence against MS — in AFib it is simply abolished by loss of the atrial kick.

AFib (quivering left atrium)

WHAT YOU SEE

The trembling bag/balloon labeled QUIVERING AFIB beside the gate.

WHAT IT ENCODES

Chronic LA pressure and stretch in MS make atrial fibrillation extremely common. AFib worsens symptoms (loss of atrial kick, rate-dependent filling) and is the key driver of stasis and thromboembolism in these patients.

THE BOARD TRAP

Overlooking that new AFib can precipitate acute decompensation in MS by shortening diastolic filling time.

Left atrial enlargement

WHAT YOU SEE

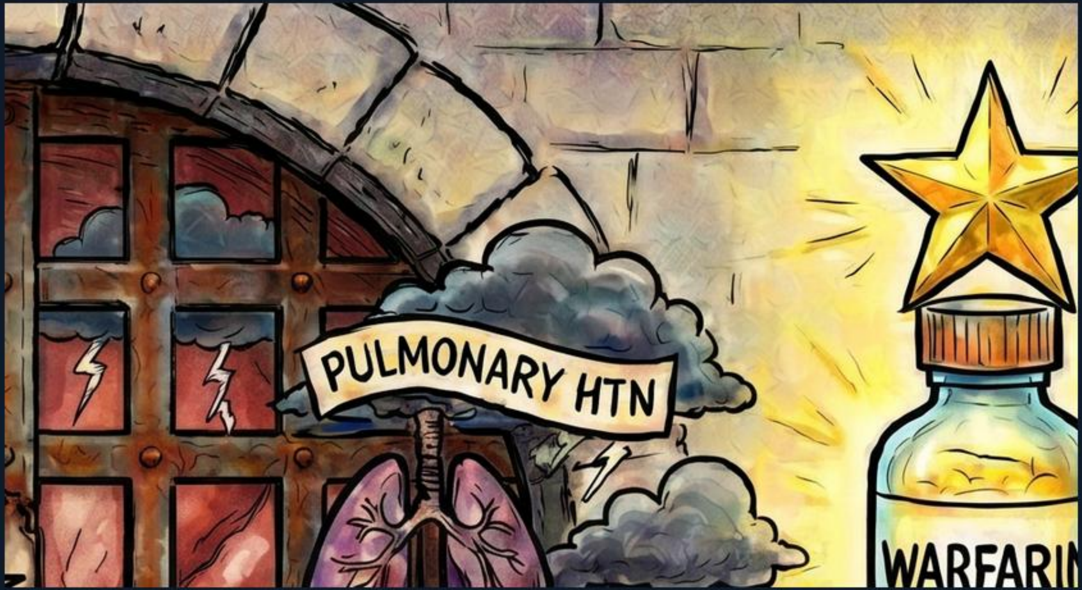
The large red bulging balloon labeled LA ENLARGEMENT pressing against the gate.

WHAT IT ENCODES

The obstructed mitral valve raises left atrial pressure and progressively dilates the LA. LA enlargement underlies AFib, can cause dysphagia or hoarseness by compression, and is the structural substrate for stasis and clot.

THE BOARD TRAP

Attributing LV enlargement to MS — the LV is typically normal/underfilled; it is the LEFT ATRIUM that enlarges.



Pulmonary hypertension

WHAT YOU SEE

A pair of lungs with a storm cloud, labeled PULMONARY HTN.

WHAT IT ENCODES

Sustained elevation of LA pressure transmits backward into the pulmonary circulation, causing pulmonary venous and then pulmonary arterial hypertension. Over time this leads to right heart strain and right-sided failure — a marker of advanced disease.

THE BOARD TRAP

Forgetting MS is a left-sided lesion that ultimately presents with RIGHT heart failure signs via pulmonary HTN.

Systemic emboli (clot cannonballs)

WHAT YOU SEE

Small iron cannonballs labeled LITTLE CLOT CANNONBALLS near the enlarged atrium.

WHAT IT ENCODES

Stasis in the dilated, fibrillating left atrium (especially the appendage) forms thrombus that can embolize to the brain and systemic arteries. Systemic embolism, including stroke, is a major complication of rheumatic MS with AFib.

THE BOARD TRAP

Thinking emboli in MS go to the lungs — left-atrial clot causes SYSTEMIC arterial emboli (stroke), not pulmonary emboli.

Warfarin — anticoagulation of choice

WHAT YOU SEE

A bottle labeled WARFARIN with a gold star, INR 2-3 gauge above it.

WHAT IT ENCODES

For atrial fibrillation in rheumatic MS, warfarin is the only approved anticoagulant, with a target INR of 2 to 3. This is the classic 'valvular AFib' indication where vitamin K antagonism is mandated for stroke prevention.

THE BOARD TRAP

Choosing a DOAC for rheumatic-MS AFib — only WARFARIN (INR 2-3) is approved; DOACs are contraindicated here.

DOACs not approved

WHAT YOU SEE

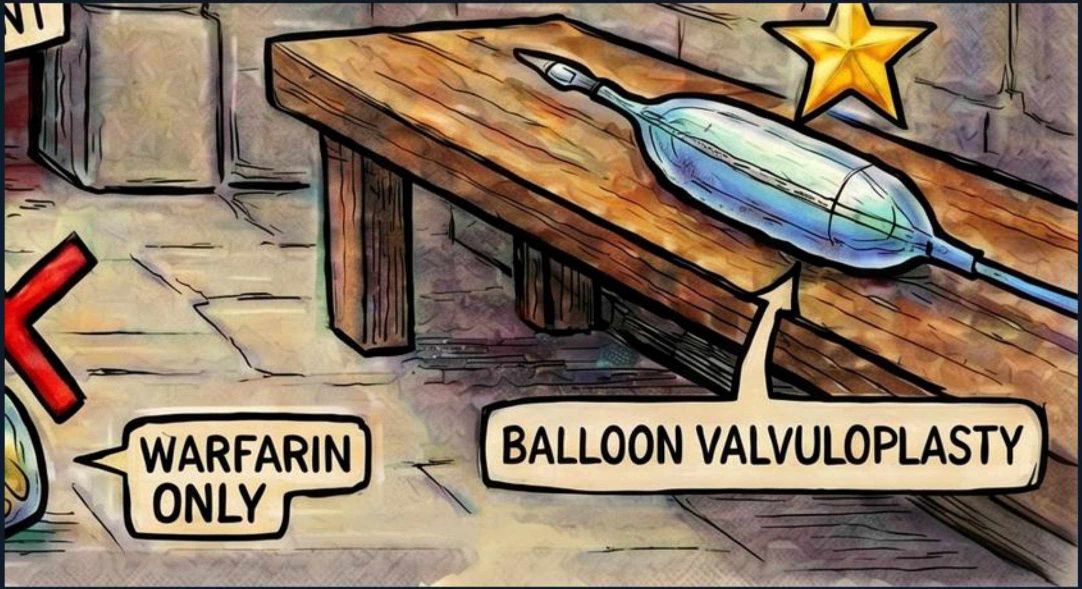
A pill bottle with a big red X, captioned DOAC / NOT APPROVED HERE (repeated bottom center as DOAC X, WARFARIN ONLY).

WHAT IT ENCODES

Direct oral anticoagulants have not been shown safe/effective for AFib in rheumatic mitral stenosis and are not approved in this setting. The exam wants you to reflexively reject a DOAC and reach for warfarin.

THE BOARD TRAP

Prescribing apixaban/rivaroxaban/dabigatran for AFib in rheumatic MS — this is the most tested wrong answer; use warfarin only.



Balloon mitral valvuloplasty

WHAT YOU SEE

A balloon catheter device on the bench with a gold star, labeled BALLOON VALVULOPLASTY.

WHAT IT ENCODES

For severe symptomatic MS with suitable valve anatomy (pliable, non-calcified, minimal MR, no LA thrombus), percutaneous balloon mitral valvuloplasty splits the fused commissures and is the preferred intervention. Unfavorable anatomy or significant MR pushes toward surgical valve replacement.

THE BOARD TRAP

Offering valvuloplasty regardless of anatomy — it requires a pliable valve, little MR, and no LA clot; otherwise choose surgery.

Summary — every symbol

1. Rheumatic / commissural fusion
Rheumatic heart disease is the dominant cause of mitral stenosis worldwide

2. Loud S1
In MS the mitral leaflets stay wide open until forced shut, so closure is abrupt and S1 is loud — as long as the leaflets remain pliable

3. Opening snap
After S2, the stenotic mitral valve domes open with a high-pitched OPENING SNAP early in diastole

4. A2–OS interval: shorter = worse
The interval between A2 and the opening snap is the single best auscultatory severity clue

5. Diastolic rumble at apex
The murmur of MS is a low-pitched mid-diastolic rumble heard best at the apex with the bell, patient in the left lateral decubitus position

6. Presystolic accentuation — sinus only
In sinus rhythm, atrial contraction pushes extra blood across the valve just before S1, producing a presystolic accentuation of the rumble

7. Presystolic murmur lost in AFib
When MS patients go into atrial fibrillation, the atrial kick is gone, so the presystolic accentuation of the murmur disappears

8. AFib (quivering left atrium)
Chronic LA pressure and stretch in MS make atrial fibrillation extremely common

9. Left atrial enlargement
The obstructed mitral valve raises left atrial pressure and progressively dilates the LA

10. Pulmonary hypertension
Sustained elevation of LA pressure transmits backward into the pulmonary circulation, causing pulmonary venous and then pulmonary arterial hypertension

11. Systemic emboli (clot cannonballs)
Stasis in the dilated, fibrillating left atrium (especially the appendage) forms thrombus that can embolize to the brain and systemic arteries

12. Warfarin — anticoagulation of choice
For atrial fibrillation in rheumatic MS, warfarin is the only approved anticoagulant, with a target INR of 2 to 3

13. DOACs not approved
Direct oral anticoagulants have not been shown safe/effective for AFib in rheumatic mitral stenosis and are not approved in this setting

14. Balloon mitral valvuloplasty
For severe symptomatic MS with suitable valve anatomy (pliable, non-calcified, minimal MR, no LA thrombus), percutaneous balloon mitral valvuloplasty splits the fused commissures and is the preferred intervention